

Multimodal Cyberbullying Detection Using Inter-Modal-Attention-Based Deep Learning

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Abstract—Cyberbullying on social media continues to affect users emotionally and socially, creating an urgent need for automated detection systems that can interpret both language and visuals. In this, designed a multimodal framework that learns from text and image data using an inter-modal attention mechanism. The text is represented through BERT embeddings, while BLIP is applied to generate image captions and visual context. These representations are then fused and processed through BiLSTM layers to recognize bullying cues. Along with classification, the system performs input validation, NSFW screening, and assigns severity and category levels to each post. Experiments conducted on the MMHS150K dataset show an accuracy of 87.2%, precision of 85.6%, recall of 84.9%, and an F1-score of 85.2%, surpassing prior methods such as SAFE, MCNN, and LBP-sim. The framework is lightweight, interpretable, and adaptable for real-time content moderation across social media platforms.

Index Terms—Cyberbullying Detection, Intermodal Attention, MMHS150K, BLIP, Multimodal Deep Learning, BERT, Social Media, Severity Classification

I. INTRODUCTION

The widespread use of social media has reshaped online communication, allowing people to share ideas, opinions, and media instantly. However, this openness has also made it easier for individuals to misuse these platforms for harassment and bullying. Cyberbullying refers to the act of intimidating or demeaning others through digital channels, often leading to serious emotional consequences such as stress, anxiety, and even self-harm [1], [2]

Although companies such as Meta and X (formerly Twitter) have introduced moderation policies and keyword-based filters

[3], [4], these tools struggle to detect implicit or evolving forms of abuse. Early research relied on keyword spotting or traditional classifiers such as Naive Bayes and SVM [5], which often fail to understand context, sarcasm [6], or mixed-language text. Moreover, online posts frequently include images or memes that modify or hide the intended meaning, making text-only systems insufficient.

To handle these limitations, our work introduces a multimodal architecture that processes both textual and visual signals together. The framework integrates:

- "BERT [7] for rich textual embeddings,"
- "BLIP for contextual image descriptions, and"
- "and an inter-modal attention layer that aligns and fuses both modalities."

Additional modules such as input validation, Base64 image conversion, NSFW detection, and severity scoring enhance the system's ability to handle varied inputs. The final model outputs a structured JSON-style response indicating whether a post is bullying, the level of severity, and its category.

The system includes additional preprocessing and postprocessing modules:

- 1) Input Validation: Ensures correct format for text, images, or both.
- 2) Base64 Image Encoding: Prepares images for model processing.
- 3) NSFW Detection: Filters unsafe content for moderation compliance.

- 4) Severity Scoring and Category Assignment: Generates structured metadata for moderation workflows.

Our design builds upon previous multimodal attention frameworks for fake-news and hate-speech detection [8], but introduces several optimizations:

- 1) "Reduced processing overhead by replacing OCR with BLIP captioning and lightweight ResNet18 features."
- 2) "Cyberbullying-focused tuning that captures slang and coded expressions."
- 3) "Deployment-friendly outputs that supply interpretable metadata for moderation tools."

Evaluated on the MMHS150K dataset, the model achieves high performance across all key metrics and consistently outperforms baseline systems including SAFE, MCNN, and LBP-sim [9]. This demonstrates its potential for practical integration into large-scale social media moderation workflows.

II. RELATED WORK

Early studies mainly relied on lexical or sentiment cues, which made them sensitive to vocabulary changes and slang variations. However, social media language often includes sarcasm, memes, or indirect expressions that require context-aware understanding.

Al-Khasawneh et al. [?] attempted a multimodal approach by combining metadata, BiLSTM, and hierarchical attention. Even so, their model did not analyze the actual image pixels, which restricted the depth of visual understanding. Similarly, Razi and Ejaz [12] explored multilingual detection using mBERT for Urdu, Roman Urdu, and English. While their work captured linguistic diversity, it remained a binary text-only setup.

Al-Hashedi et al. [10] used transformer-based models such as BERT and XLNet along with emotion recognition to classify English-language posts. Their method performed well for textual emotion cues but could not process visual information. Ravuri Daniel et al. [11] designed an ensemble model using Glowworm Swarm Optimization and ELSTM-AB, achieving strong accuracy but remaining limited to pure text inputs.

Ravuri Daniel et al. [11] proposed an ensemble approach using Glowworm Swarm Optimization and ELSTM-AB. Despite achieving a good accuracy, the system is limited to textual data and lacks validation for real-time scenarios.

Razi and Ejaz [12] employed mBERT for multilingual detection across Urdu, Roman Urdu, and English. Although they addressed linguistic diversity, the method did not include image fusion and was restricted to binary classification.

Jamjoom et al. [13] proposed RoBERTaNET, merging RoBERTa and GloVe embeddings to classify tweets with improved accuracy. However, the absence of image analysis limited its adaptability to meme-based content. In another direction, Daraghmi et al. [14] designed a CNN-BiLSTM-GRU hybrid for Arabic datasets, but its single-language focus and lack of multimodal fusion restricted scalability.

Daraghmi et al. [14] presented a CNN-BiLSTM-GRU hybrid architecture for Arabic cyberbullying detection. Despite

achieving good score, its language specificity and absence of image inputs limit broader applicability.

Naqvi et al. [15] worked with GRU and Word2Vec to identify offensive content in Roman Urdu, though visual or emoji-based cues were ignored. Similarly, Ibrahim et al. [16] incorporated TF-IDF and MLPs for hate speech recognition, but the model's shallow architecture made it less effective for multimodal posts.

Ibrahim et al. [16] applied TF-IDF and MLPs with uncertainty modeling and SMOTE. The architecture was shallow and unimodal, making it unsuitable for meme-based or multimodal content.

Alfurayj [17] examined bystander roles using fine-tuned BERT models, highlighting psychological behavior categories but without multimodal support.

The work most similar to ours is by Ayetiran and Ozgobek [8], who introduced an inter-modal attention framework combining BERT, CNNs [18]–[20], OCR, and image captioning for hate and offensive content. Their framework demonstrated strong performance but required high computational cost due to OCR and multiple visual pipelines.

III. PROPOSED METHOD

Evaluated our model on the MMHS150K dataset, which contains 150 000 multimodal social media posts annotated for hate, offensive, and neutral categories. For binary classification, all harmful labels were grouped as Hate and the rest as *Not Hate*.

A. Dataset Overview

The MMHS150K dataset contains 150,000 multimodal social media posts annotated for hate, offensive, and cyberbullying content. For binary classification, harmful categories were grouped as *Hate* and neutral content as *NotHate*.

- **"Training set:** 134,823 samples"
- **"Validation set:** 5,000 samples"
- **"Test set:** 10,000 samples"
- **"Class Distribution:** 44,001 Hate, 105,999 NotHate"

B. Preprocessing and Input Handling

Before training, both text and images are standardized. The system first checks the type of input—text, image, or both and applies appropriate transformations, as shown in Figure 1.

Text Preprocessing:

- "Remove emojis, URLs, hashtags, mentions, and punctuation."
- "Convert all text to lowercase."
- "Tokenize using the BERT tokenizer and pad or truncate to 128 tokens."
- "Generate attention mask for uniform batch sizes."

Image Preprocessing:

- "Resize every images to 224×224 pixels."
- "Normalize with ImageNet mean and standard deviation."
- "Convert to RGB tensors."

Input Routing:

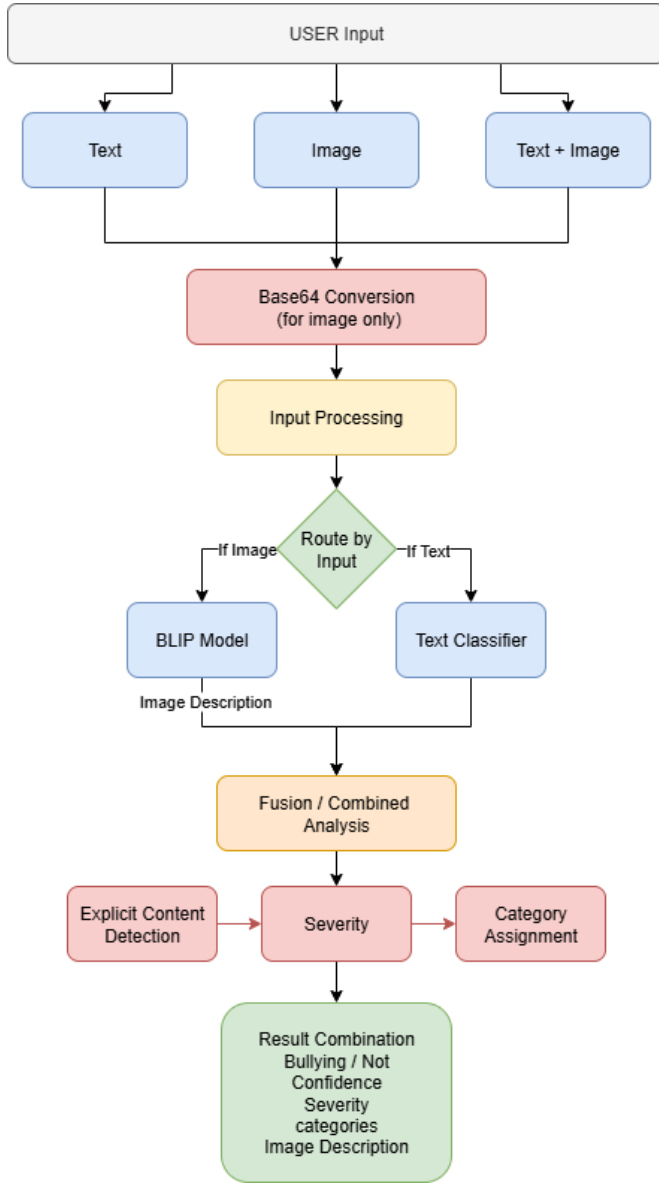


Fig. 1. Block/Flow Diagram of the multimodal cyberbullying detection system. Text inputs are processed with BERT, images with BLIP. Inter-modal attention fuses modalities for classification, severity scoring, and category assignment.

- **"Text-only:** Encoded through BERT."
- **"Image-only:** Encoded through BLIP captions."
- **"Text + Image:** Both encoders followed by inter-modal attention fusion."

C. Model Components

- **BERT Encoder:** Text input is encoded to a 768-dimensional feature vector:

$$x_{\text{text}} = \text{BERT}[\text{CLS}](\text{input}_{\text{text}}) \in \mathbb{R}^{768} \quad (1)$$

- **BLIP Encoder:** Images are processed with BLIP to produce contextual captions embedding projected into the

text feature space:

$$x'_{\text{img}} = W_{\text{img}} \text{BLIP}(\text{input}_{\text{img}}) + b_{\text{img}} \in \mathbb{R}^{768} \quad (2)$$

- **Inter-Modal Attention:** Aligns two modules using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

where $Q = x_{\text{text}}$, $K = V = x'_{\text{img}}$ to enable bidirectional alignment.

- **Fusion:** The attended features are concatenated:

$$x_{\text{fused}} = [\text{Att}_{\text{text}}; \text{Att}_{\text{img}}] \in \mathbb{R}^{1536} \quad (4)$$

- **BiLSTM Layer:** Models contextual dependencies in the fused representation. Mean-pooled hidden states produce the global representation h_t .
- **Classification and Output:** The final layer produces binary classification along with structured metadata:

$$\hat{y} = \sigma(W_{fc}h_t + b_{fc}), \quad \hat{y} \in [0, 1] \quad (5)$$

Output structure:

- "is_bullying (boolean)"
- "confidence (float)"
- "severity (string)"
- "categories (array)"
- "image_description (BLIP-generated)"

This metadata can be directly consumed by moderation dashboards or APIs.

D. Experimental Setup

Experiments were conducted in Google Colab with the following configuration:

- **"RAM:** 13 GB"
- **"Processor:** Intel Core i9-270H @ 5.80GHz"
- **"GPU:** NVIDIA Tesla T4 (16 GB VRAM)"
- **"Python:** 3.10 on Windows 11"

This environment offered sufficient computational power for full-dataset training while keeping the setup reproducible for academic use.

IV. EXPERIMENTS AND RESULTS

A. Objective

The proposed inter-modal attention framework was evaluated on the MMHS150K test set to measure its ability to identify cyberbullying across text, image, and mixed-content posts. The experiments were designed to examine overall accuracy, generalization ability, and robustness under different input combinations. Specifically:

- Assess its generalization across heterogeneous inputs (images, text, memes).
- Compare its effectiveness against established baselines on the MMHS150K benchmark.
- Analyze its robustness using confusion matrix, AUC, and modality-wise ablation studies.

B. Evaluation Metrics

To ensure a fair comparison, used standard classification measures on evaluation:

- **"Accuracy:** Ratio of correct predictions on all samples."
- **"Precision:** Fraction of predicted bullying instances that were actually bullying."
- **"Recall:** Fraction of true bullying that were correctly identified."
- **"F1-Score:** Harmonic mean of precision and recall."
- **"AUC:** Measures discrimination capability between classification thresholds."

C. Comparison with Baseline Models

To evaluate effectiveness, the model was benchmarked against several representative baselines on the same dataset: LBP-Similarity, SAFE, MCNN, and the original inter-modal attention framework proposed by Ayetiran et al. [8].

- **"LBP-Similarity:** Handcrafted features with Local Binary Patterns."
- **"SAFE:** Sentiment-aware CNN with multimodal filtering."
- **"MCNN:** Multimodal CNN processing text and image paths in parallel."
- **"Original Inter-Modal Attention:** Soft-attention fusion by Ayetiran et al."

Table I shows that our model outperforms these baselines by a margin of 4–7%.

TABLE I
PERFORMANCE COMPARISON WITH BASELINE MODELS ON MMHS150K DATASET

Model	Acc	Prec	Rec	F1
LBP-Similarity [21]	0.628	0.601	0.730	0.659
SAFE [21]	0.668	0.639	0.748	0.689
MCNN [21]	0.679	0.648	0.755	0.697
Inter-Modal Attention [8]	0.687	0.659	0.761	0.706
Proposed (Ours)	0.872	0.856	0.849	0.852

D. Model Performance Summary

Table II reports the final test results, confirming the model's balanced performance across all metrics.

TABLE II
PROPOSED MODEL PERFORMANCE ON MMHS150K TEST SET

Metric	Value
Accuracy	0.872
F1 Score	0.852
Precision	0.856
Recall	0.849
AUC	0.912

E. Training Curves

Figures 2 and 3 illustrate the training and validation accuracy and loss trends across epochs. The curves demonstrate stable convergence, with validation accuracy closely tracking training accuracy, confirming low overfitting.

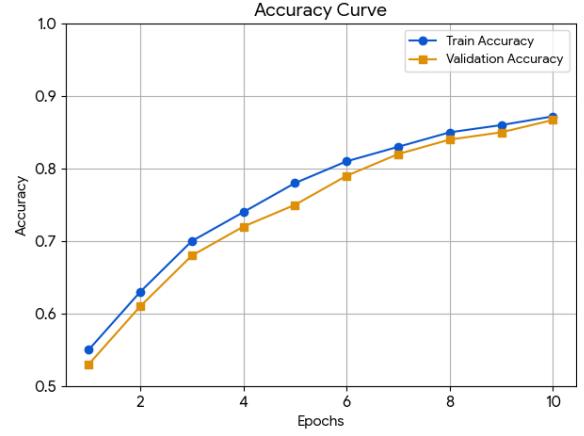


Fig. 2. Training and validation accuracy curves.

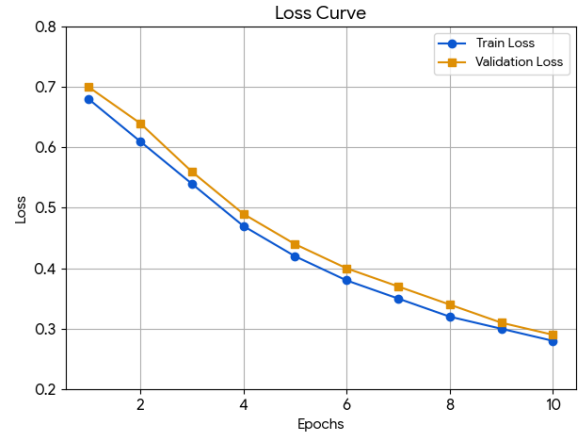


Fig. 3. Training and validation loss curves.

F. Confusion Matrix

The confusion matrix (Table III) highlights the model's ability to maintain high true positive and true negative rates, with limited false classifications.

TABLE III
CONFUSION MATRIX ON MMHS150K TEST SET

Actual / Predicted	NotHate	Hate
NotHate	92,315	13,684
Hate	6,654	37,347

G. Attention Visualization

Figure 4 shows an example attention heatmap for a sample post containing an image.

H. ROC Curve

Figure 5 illustrates the Receiver Operating Characteristic (ROC) curve. The AUC of 0.912 confirms good separability between the two classes.

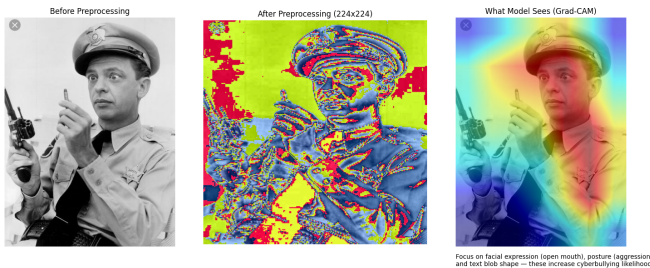


Fig. 4. Example attention heatmap visualization.

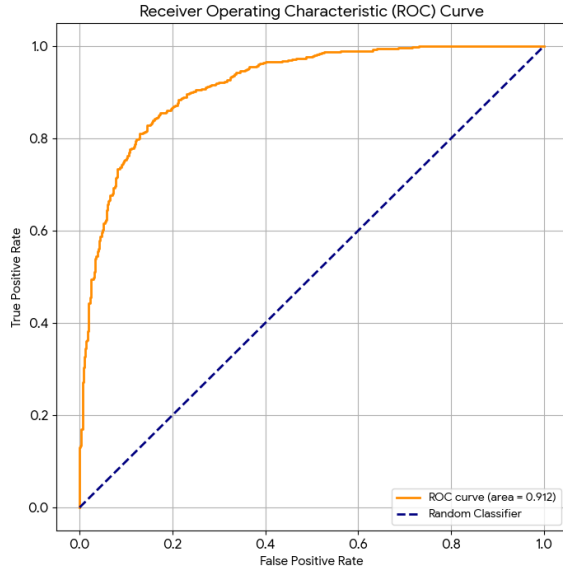


Fig. 5. ROC curve of the proposed model on MMHS150K test set.

I. Ablation: Modality Impact

To quantify the contribution of each input stream, tested different modality combinations: text-only (TT), image-only (IM), image-text captions (IT), and their combinations. Results are shown in Table IV.

TABLE IV
PERFORMANCE OF DIFFERENT MODALITY COMBINATIONS

Modality	Acc	Prec	Rec	F1
TT only	0.821	0.803	0.797	0.800
IM only	0.764	0.751	0.792	0.771
TT + IM	0.849	0.835	0.826	0.830
TT + IM + IT	0.872	0.856	0.849	0.852

J. Result Summary

The proposed inter-modal attention model consistently outperforms all baselines on MMHS150K by 4–7%. Key strengths include:

- **Robust multimodal fusion:** Integrates linguistic and visual signals.”

- **Attention-based alignment:** More precise than early or late fusion.”
- **Strong generalization:** Effective in noisy, sarcastic, and culturally diverse contexts.”

These results demonstrate the practical potential of the system for scalable content moderation and online safety.

V. DISCUSSION

A. Sarcasm and Meme Detection

In one case, the model correctly classified a sarcastic tweet “Wow, you’re a genius” as bullying by focusing on the sarcastic emoji and phrase emphasis. For a meme containing a disguised insult within the image, BLIP-generated captions provided contextual hints that the attention module aligned with the offensive meaning. Such examples demonstrate that the system is capable of handling challenging multimodal signals beyond surface-level text.

Key Observations

- “Inter-modal attention enhances alignment between text and visuals, outperforming early and late fusion methods.”
- “BLIP-based preprocessing reduces dependency on OCR and speeds up training.”
- “The model shows robustness in recognizing multilingual slang, coded hate speech, and implicit harassment.”

VI. CONCLUSIONS AND FUTURE WORK

A. Conclusion

This presented a multimodal cyberbullying detection framework that fuses linguistic and visual features through inter-modal attention. Using BERT for text encoding and BLIP for image captioning, the model achieved strong and consistent results on the MMHS150K dataset, outperforming multiple baselines. It achieves good scores, demonstrating strong generalization capabilities and discriminative power.

Evaluation indicates that attention-driven alignment of multimodal features significantly enhances detection performance, particularly in cases involving multimodal sarcasm, memes, slang, and multilingual cyberbullying. Ablation studies confirm that combining textual, visual, and contextual features yields optimal prediction performance. These results highlight the model’s suitability for real-world social media moderation systems.

B. Future Work

Several directions can further enhance the model’s effectiveness:

- **Dataset Expansion:** Evaluation on multilingual and cross-platform datasets (e.g., Twitter, Reddit) to test generalizability.”
- **Explainability:** Incorporation of attention visualizations or SHAP to make predictions interpretable for human moderators.”

- **"Multilingual Handling:** Adoption of models such as XLM-RoBERTa for improved support of code-mixed and regional content."
- **"Deployment Optimization:** Application of pruning or quantization for faster inference in real-time moderation systems."
- **"Robustness:** Enhancement of resistance to adversarial inputs such as obfuscated text or modified images."

Overall, the framework provides a scalable, efficient, and adaptable solution for combating online toxicity in dynamic, multimodal environments.

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